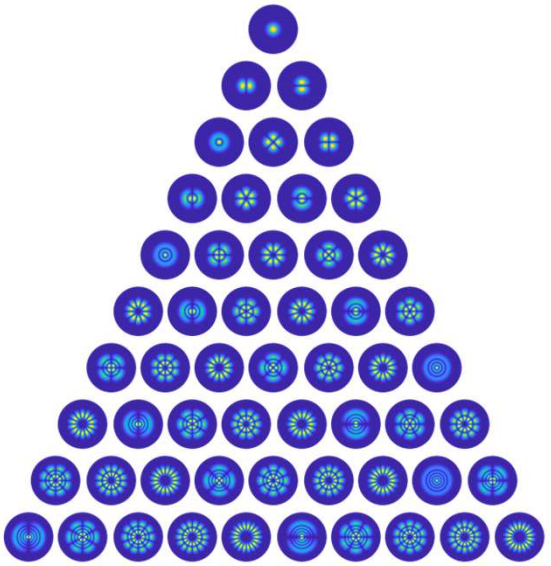
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Institute of electrical and optical communications

Advanced optical fiber communications

Wintersemester 2025/26

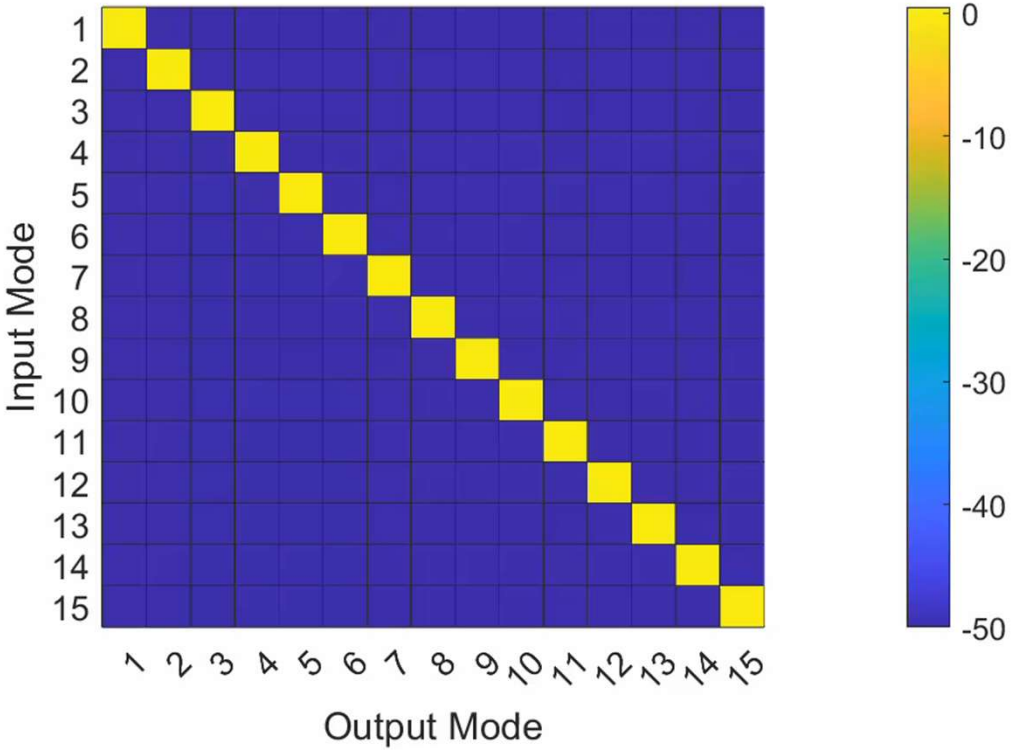
Lecture 8



Prof. Dr.-Ing. Georg Rademacher

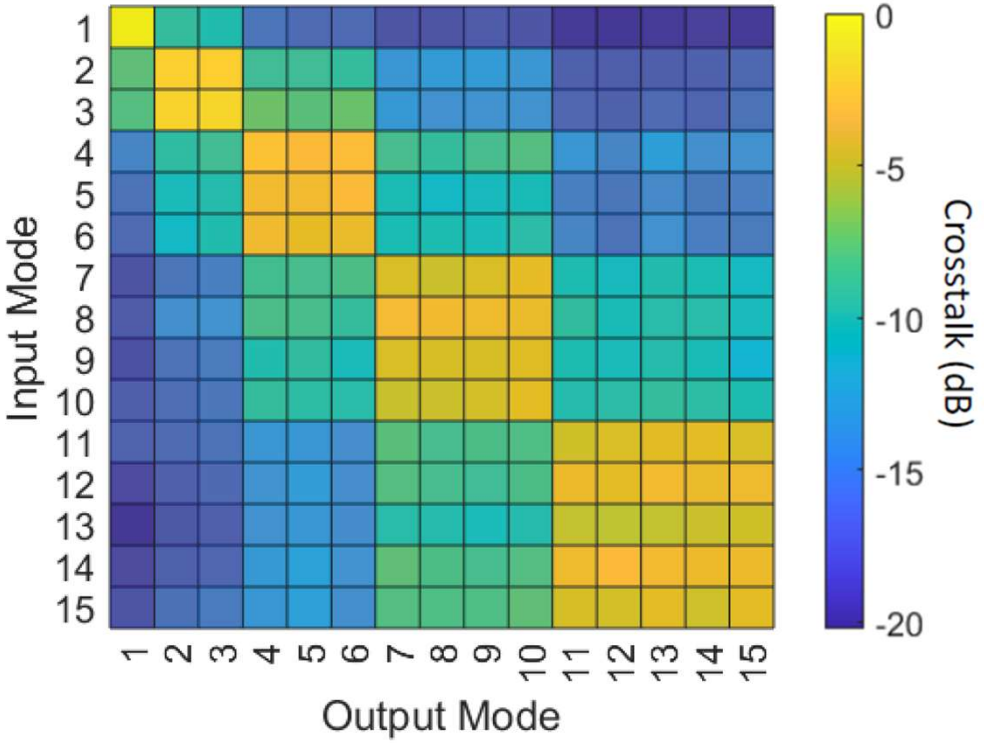
Simulation: Crosstalk vs. offset

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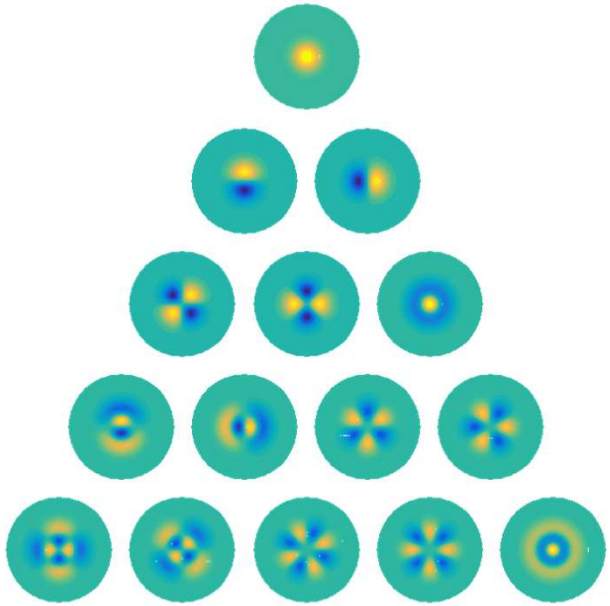




Experiment: Crosstalk



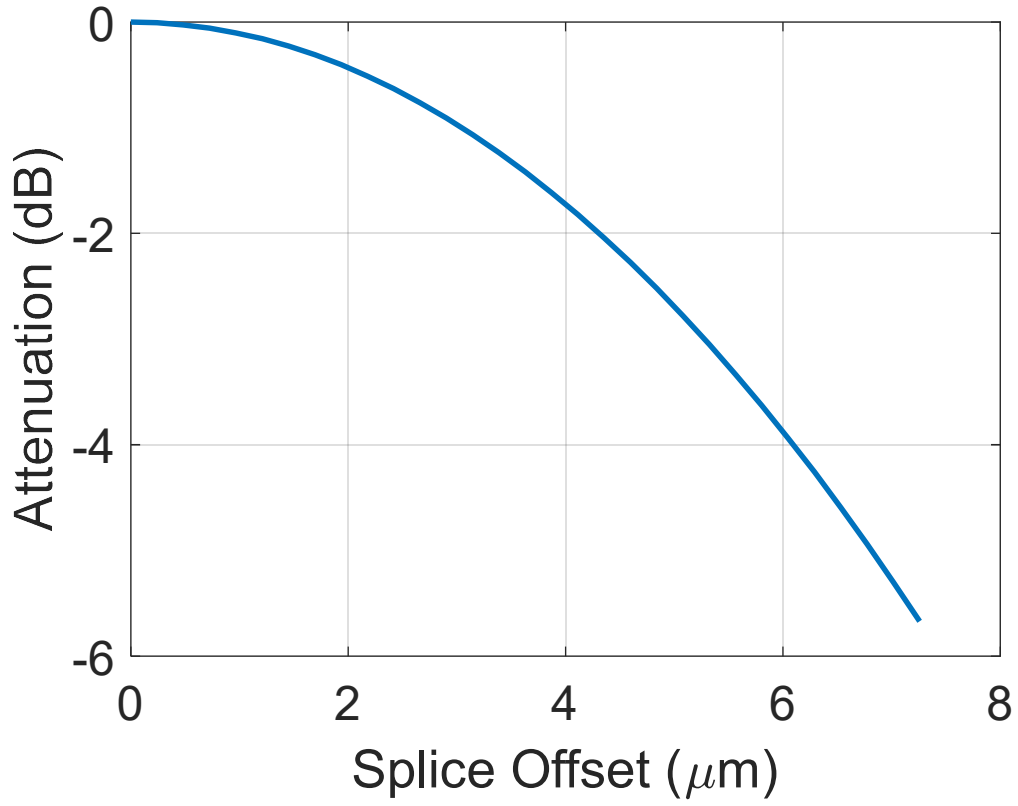
University of Stuttgart
Germany



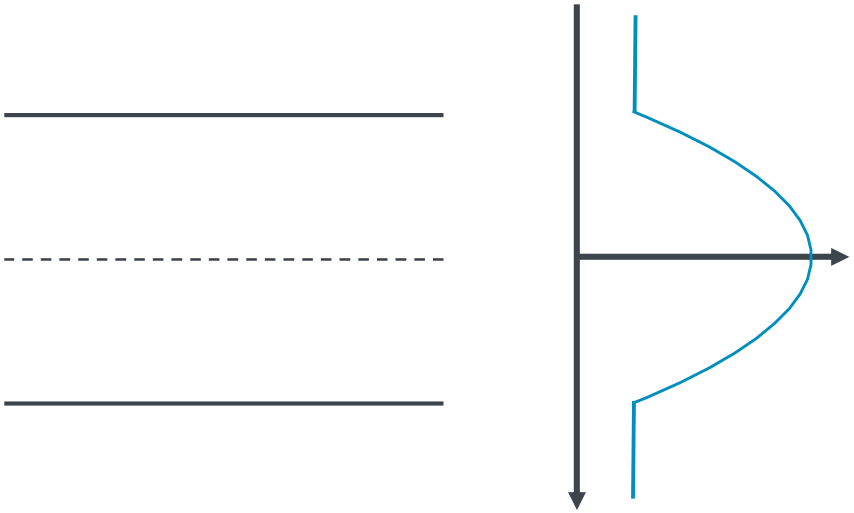
Butt-coupling of (SI) SMF

Butt-coupling of (SI) SMF

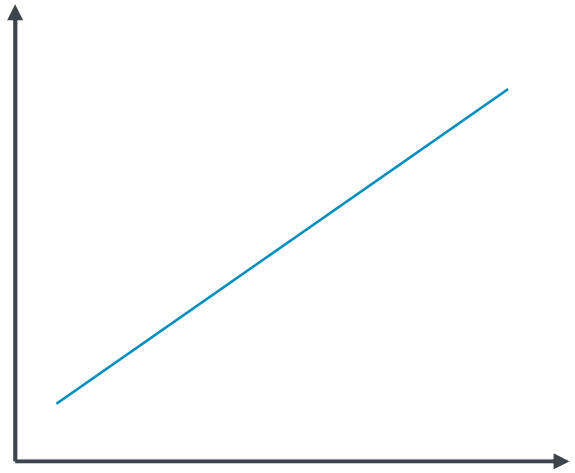
Butt-coupling of (SI) SMF



Profiledispersion



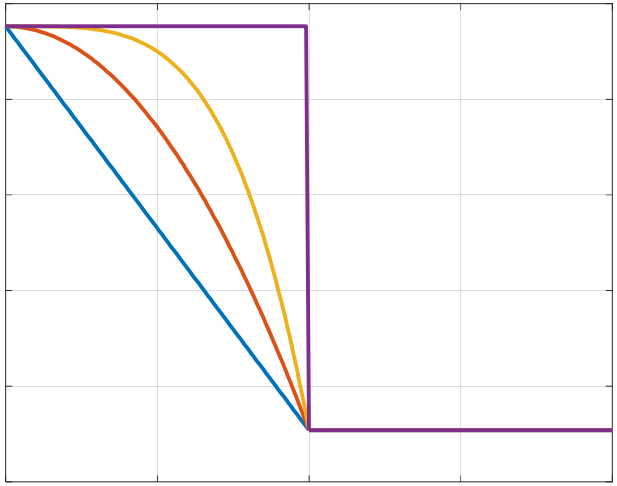
Profiledispersion



Profiledispersion

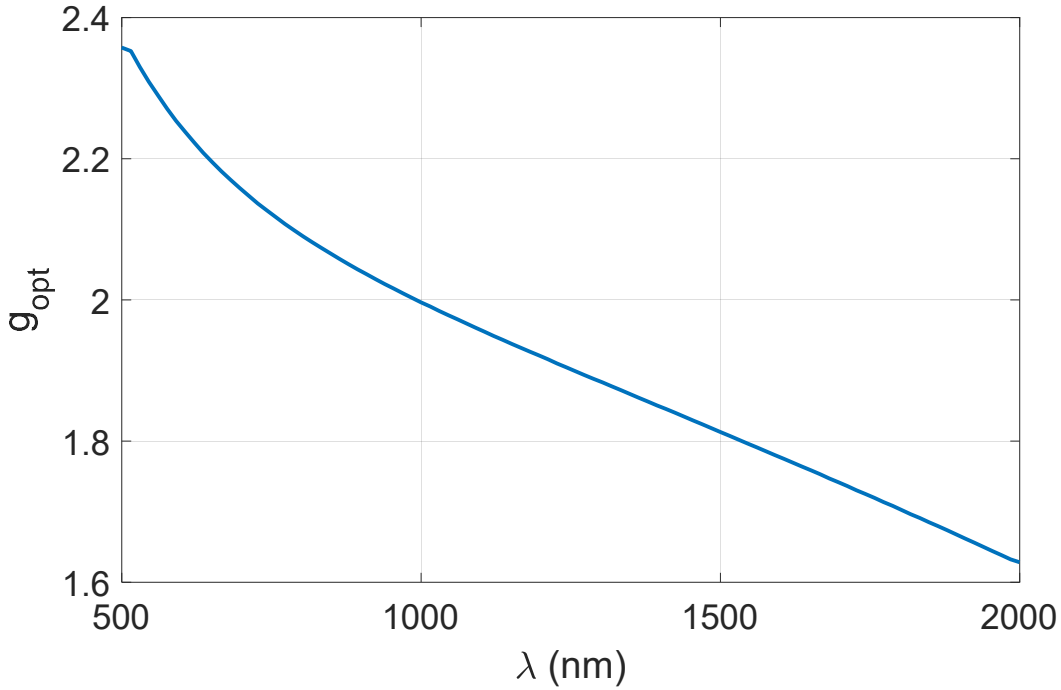
Profiledispersion

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Profiledispersion

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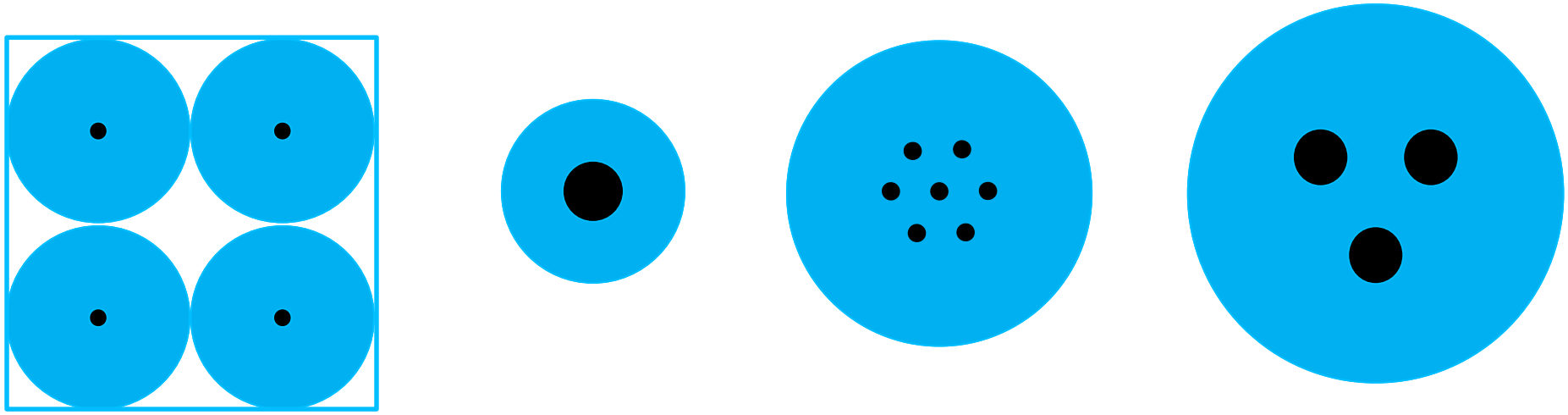


Different dispersion mechanisms

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Germany

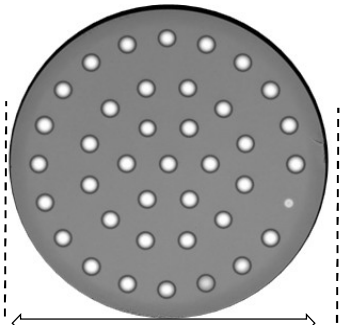
Material dispersion	Waveguide dispersion	Profile dispersion

Optical fibers for space-division multiplexing



Record SDM fiber demonstrations

13 km 38-few-mode multi-core fiber (MCF)

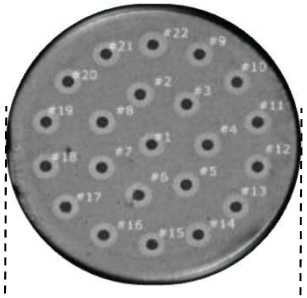


305 µm

2020
10.66 Pb/s
(C+L band)

2023
22 Pb/s
(S/C/L band)

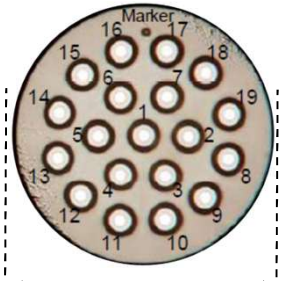
31.4 km 22-core MCF



260 µm

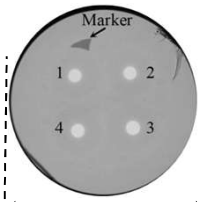
2015
2.15 Pb/s

30 km 19-core MCF



220 µm

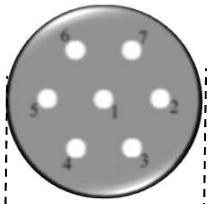
3.5 km 4-core few-mode MCF.



160 µm

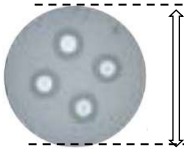
2018
1.2 Pb/s

53.7 km 7-core MCF



160 µm

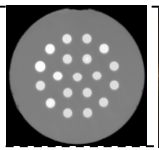
54-100 km 4-core MCF



125 µm

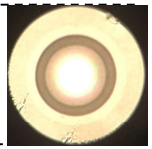
2022
1.02 Pb/s

66 km randomly coupled MCF



2023
1.7 Pb/s

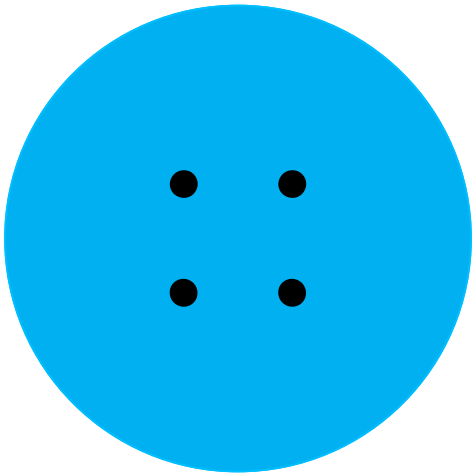
26 km 55-mode fiber



2023
3.56 Pb/s

Weakly coupled Homogenous-core MCF

University of Stuttgart
Germany



Design and fabrication of ultra-low crosstalk and low-loss multi-core fiber

Tetsuya Hayashi,* Toshiki Taru, Osamu Shimakawa, Takashi Sasaki, and Eisuke Sasaoka

Optical Communications R&D Laboratories, Sumitomo Electric Industries, Ltd., 1 Taya-cho, Sakae-ku, Yokohama, 244-8588 Japan
**t-hayashi@sei.co.jp*

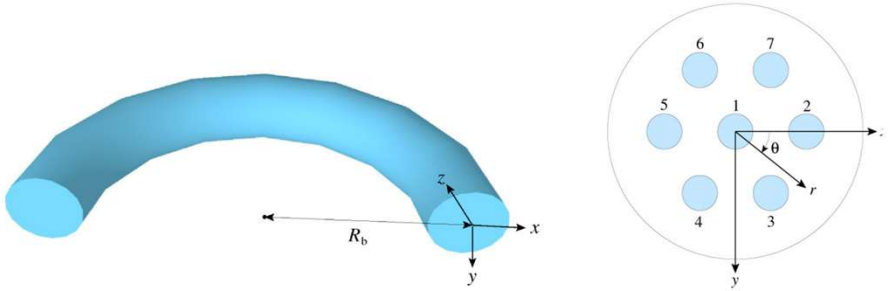
Multi-core fiber design and analysis: coupled-mode theory and coupled-power theory

Masanori Koshiba,[1,*] Kunimasa Saitoh,[1] Katsuhiro Takenaga[2], and Shoichiro Matsuo[2]

[1]Division of Media and Network Technologies, Hokkaido University, Sapporo 060-0814, Japan
[2]Optics and Electronics Laboratory, Fujikura Ltd., Sakura 285-8550, Japan
**koshiba@ist.hokudai.ac.jp*

Weakly coupled Homogenous-core MCF

University of Stuttgart
Germany



Weakly coupled Homogenous-core MCF

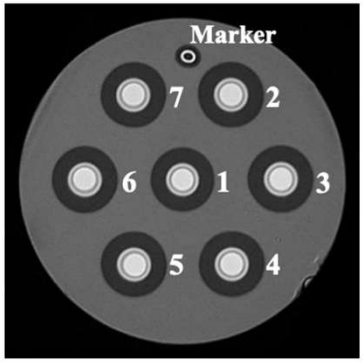


Fig. 9. Cross section of the fabricated MCF [16].

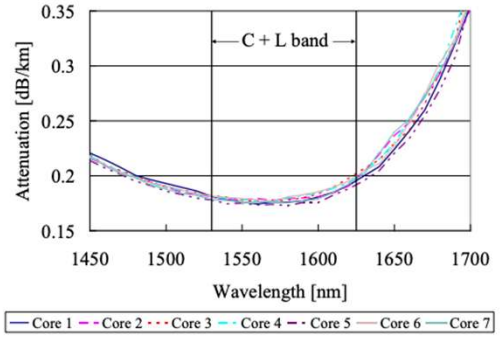


Fig. 10. Attenuation spectra of cores of the fabricated MCF.

Table 1. Designed Optical Properties of Each MCF Core at $\lambda = 1550$ nm

	Designed value
λ_{cc} [nm]	1496
MFD [μm]	9.86
A_{eff} [μm^2]	79.6
CD [ps/nm/km]	23.3
D. Slope [ps/nm^2/km]	0.063

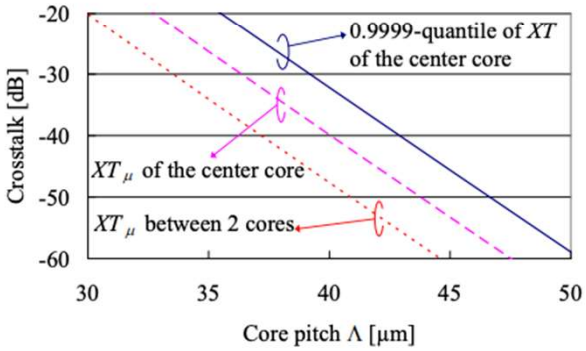
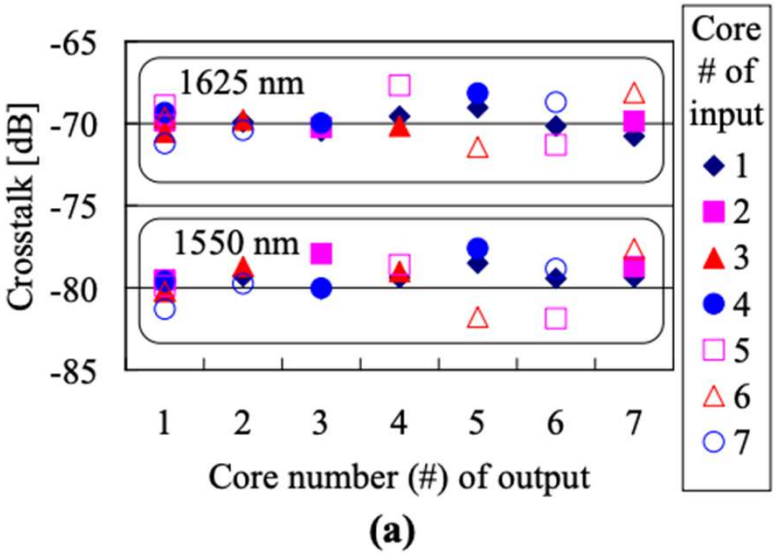


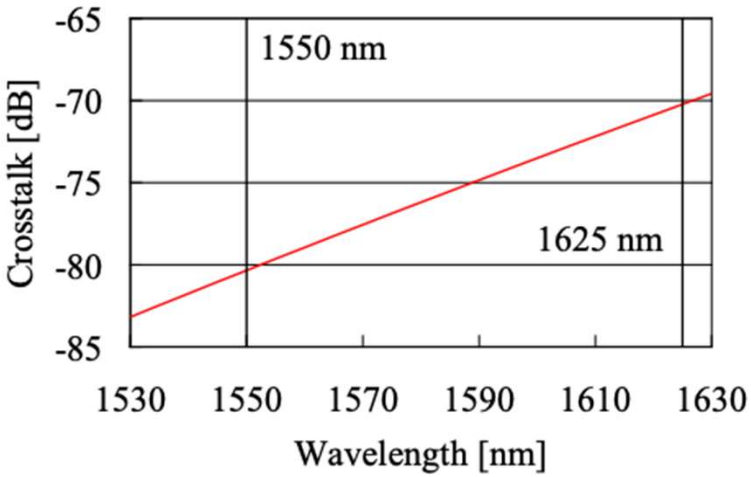
Fig. 7. Relationship between the core pitch Λ and the crosstalk for the designed core.

Weakly coupled Homogenous-core MCF

University of Stuttgart
Germany



(a)



(b)

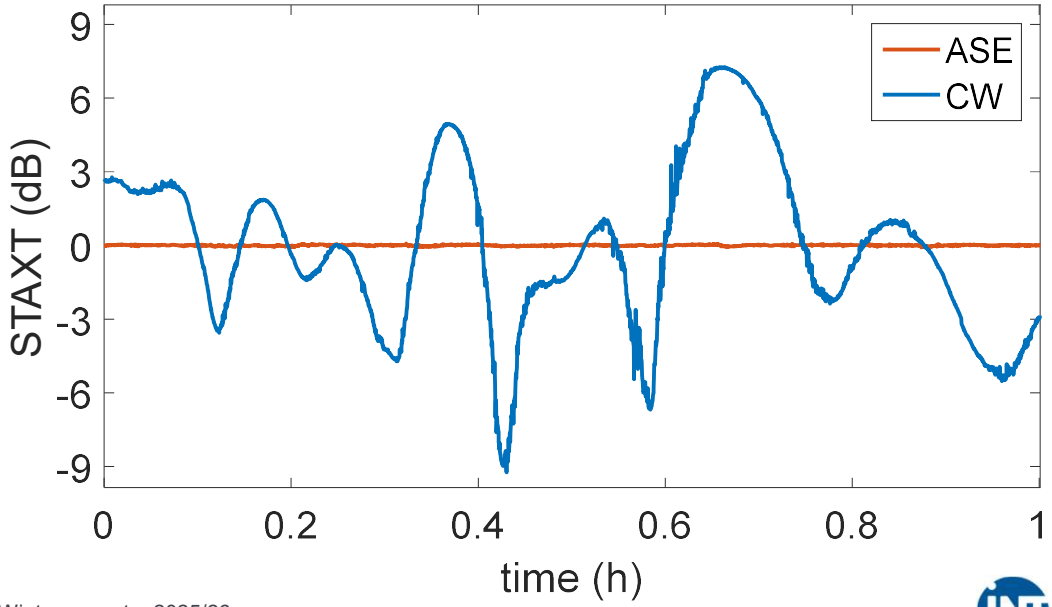
Research Article | Vol. 25, No. 10 | 15 May 2017 | OPTICS EXPRESS 12020
Optics EXPRESS

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Germany

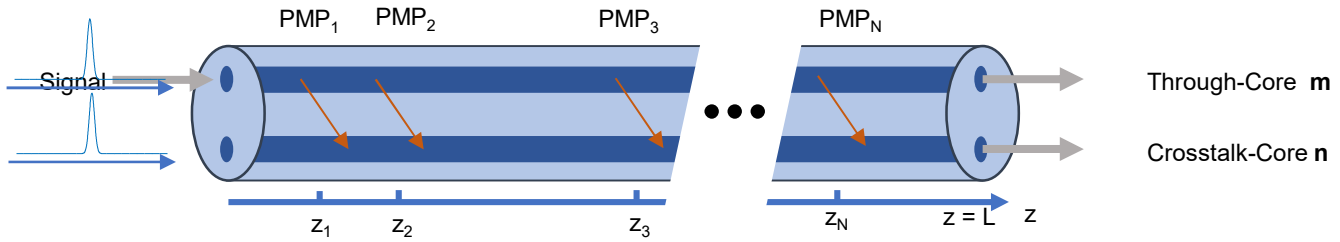
Crosstalk dynamics in multi-core fibers

GEORG RADEMACHER,* RUBEN S. LUÍS, BENJAMIN J. PUTTNAM, YOSHINARI AWAJI, AND NAOYA WADA

Photonics Systems Research Laboratory, National Institute of Information and Communications Technology (NICT), 4-2-1, Nukui-Kitamachi, Koganei, Tokyo, 184-8795, Japan
**georg.rademacher@nict.go.jp*



Modeling of inter-core crosstalk

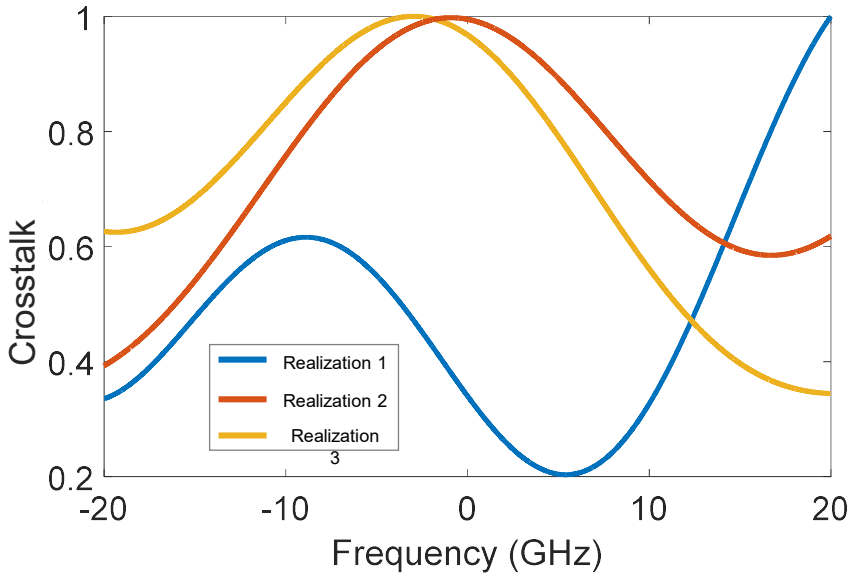


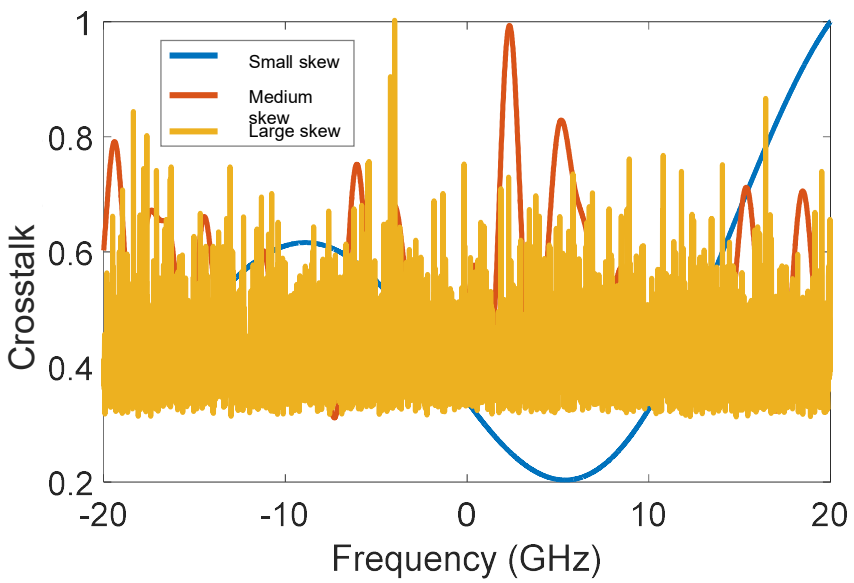
T. Hayashi et al., “Behavior of Inter-Core Crosstalk as a Noise and Its Effect on Q -Factor in Multi-Core Fiber,” IEICE Trans. Commun., 2014

R. Luís et al., “Time and Modulation Frequency Dependence of Crosstalk in Homogeneous Multi-Core Fibers,” *JLT,* 2016

Influence of random phase / inter-core skew on transfer function

$$H_{XT}(L,\omega) = -jK_{nm}\sum_{k=1}^{N} e^{-j\boldsymbol{\phi_k}} e^{-js_{nm}\boldsymbol{z_k}\omega}$$

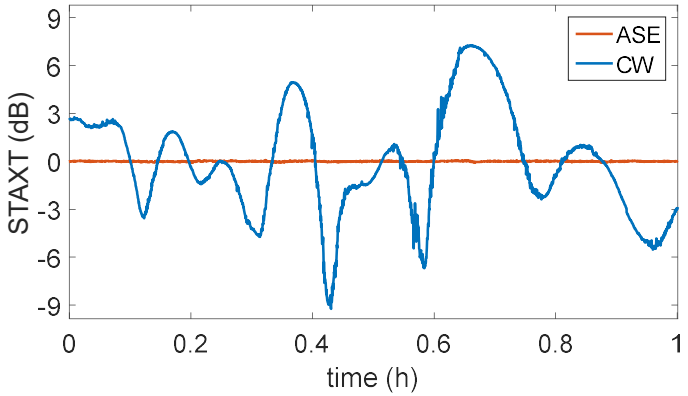




Variance of average crosstalk-power

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Germany

$$H_{XT}(L,\omega) = -jK_{nm}\sum_{k=1}^{N} e^{-j\boldsymbol{\phi_k}} e^{-js_{nm}\boldsymbol{z_k}\omega}$$

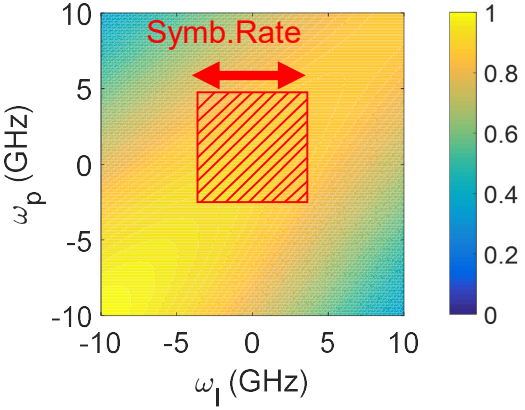


Variance of average crosstalk-power

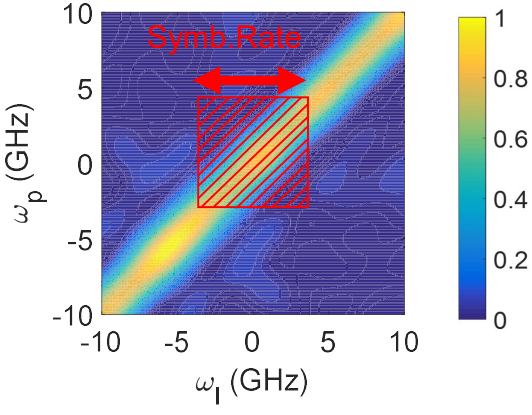
University of Stuttgart
Germany

$$\mathrm{VAR}_{\mathrm{XT}} \propto \sum_{l=1}^{M} \mathrm{VAR}(\boldsymbol{\Omega_l}) + \sum_{l \neq p}^{M} \sum_{p \neq l}^{M} \mathrm{COV}(\boldsymbol{\Omega_l}, \boldsymbol{\Omega_p})$$

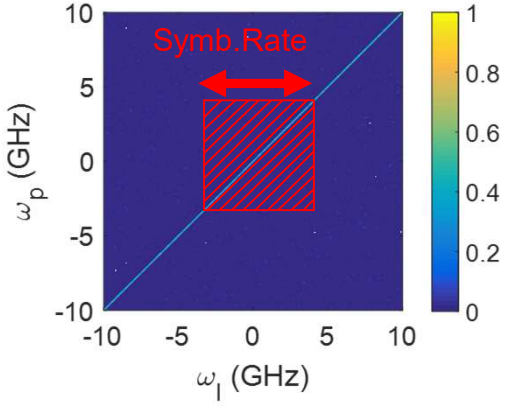
0.025 ns skew



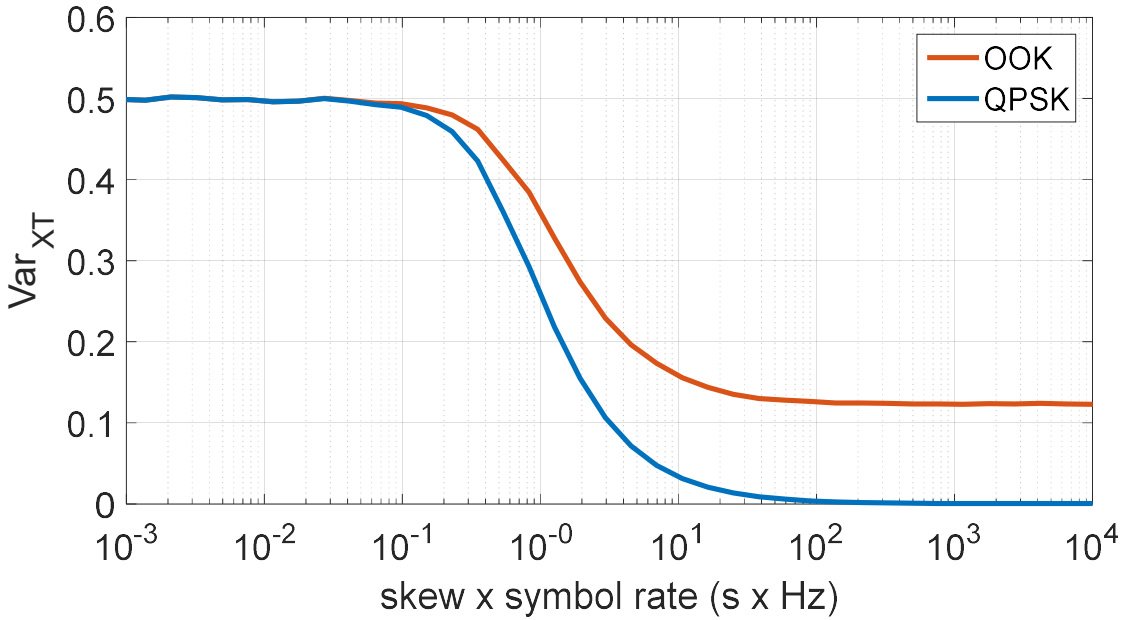
0.25 ns skew



250 ns skew



Variance of average crosstalk power





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Germany

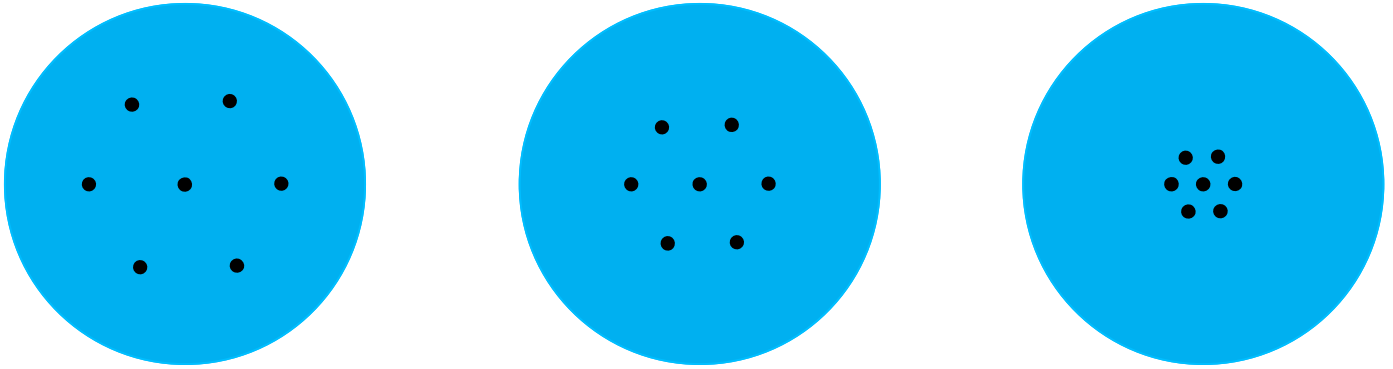
Concept Test

Randomly-Coupled Multi-Core Fiber Technology

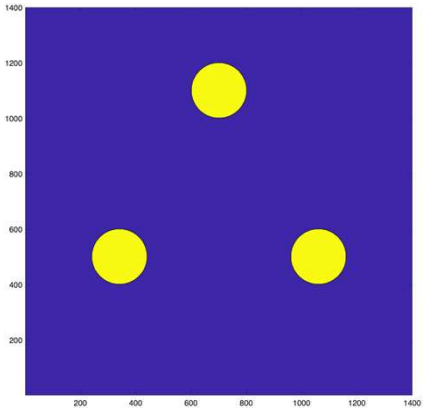
This article considers an even higher density spatially parallel integration technique that allows for random coupling between the cores of multi-core fibers.

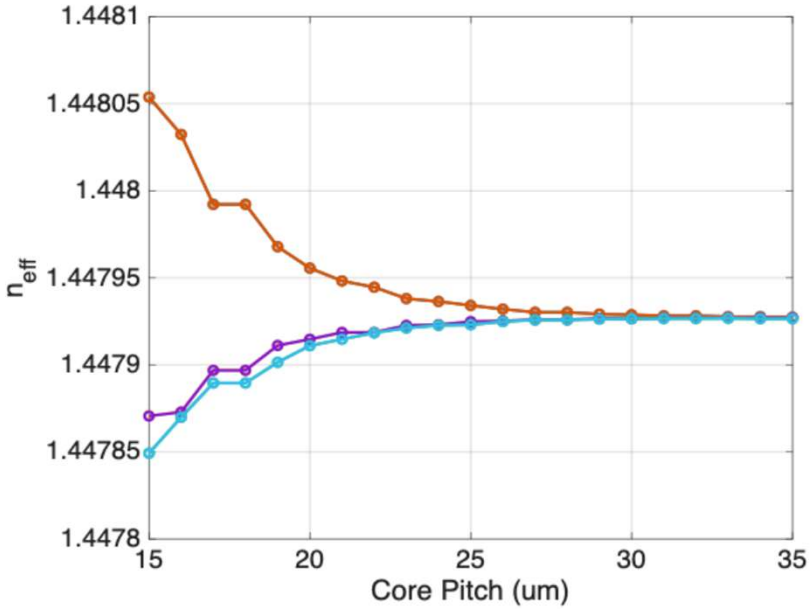
By Tetsuya Hayashi, *Senior Member IEEE*, Taiji Sakamoto, *Member IEEE*, Yusuke Yamada, Roland Ryf, *Fellow IEEE*, René-Jean Essiambre, *Fellow IEEE*, Nicolas Fontaine, *Senior Member IEEE*, Mikael Mazur, *Member IEEE*, Haoshuo Chen, *Senior Member IEEE*, and Takemi Hasegawa, *Member IEEE*

Properties for CC-MCF

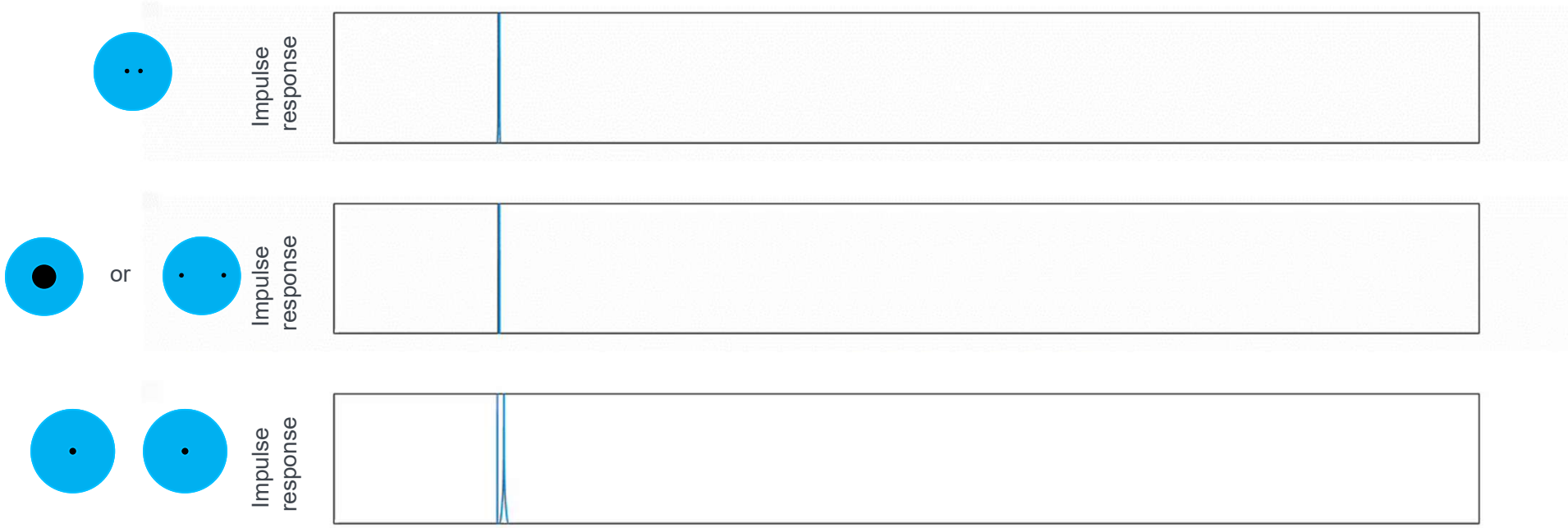


Properties for CC-MCF





Temporal spread of signals in SDM fibers

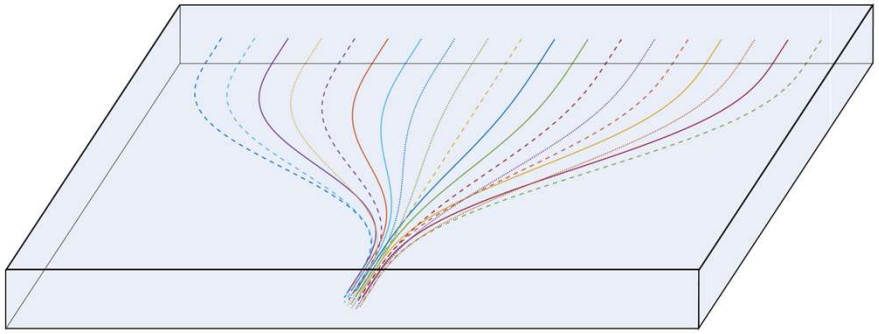


Randomly Coupled 19-Core Multi-Core Fiber with Standard Cladding Diameter

Georg Rademacher[(1)], Menno van den Hout[(1,2)], Ruben S. Luís[(1)], Benjamin J. Puttnam[(1)], Giammarco Di Sciullo[(1,3)], Tetsuya Hayashi[(4)], Ayumi Inoue[(4)], Takuji Nagashima[(4)] Simon Gross[(5)], Andrew Ross-Adams[(6)], Michael J. Withford[(6)], Jun Sakaguchi[(1)], Cristian Antonelli[(3)], Chigo Okonkwo[(2)], Hideaki Furukawa[(1)]

[(1)] NICT, 4-2-1, Nukui-Kitamachi, Koganei, Tokyo, 184-8795, Japan,
[(2)] High Capacity Optical Transmission Lab, Eindhoven University of Technology, Eindhoven, The Netherlands
[(3)] University of L'Aquila and CNIT, 67100, L'Aquila, Italy
[(4)] Sumitomo Electric Industries, Ltd., 1 Taya-cho, Sakae-ku, Yokohama 244-8588, Japan
[(5)] MQ Photonics Research Centre, School of Engineering, Macquarie Univ., Sydney, Australia.
[(6)] MQ Photonics Research Centre, School of Math. and Physical Sciences, Macquarie Univ., Sydney, Australia.
georg.rademacher@nict.go.jp

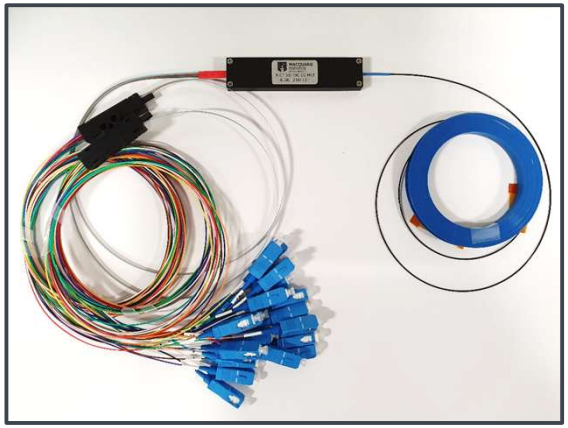
Input single mode fiber array



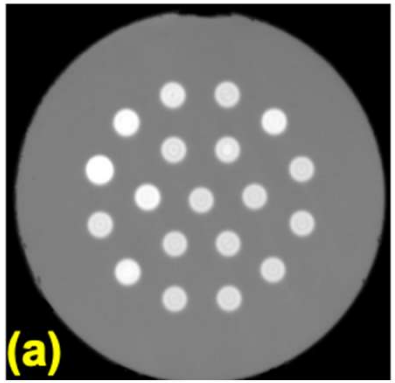
Output RC-19-MCF

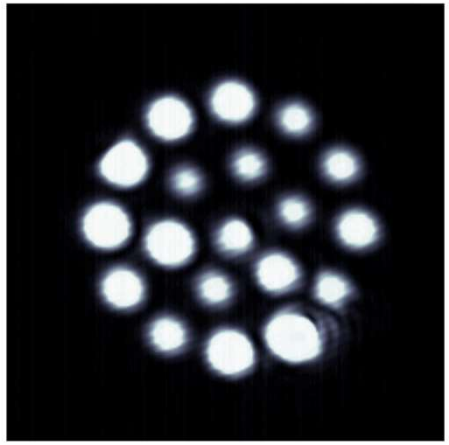
S. Gross and M.J. Withford, "Ultra fast-laser-inscribed 3D integrated photonics," Nanophotonics 4 (1) (2015)

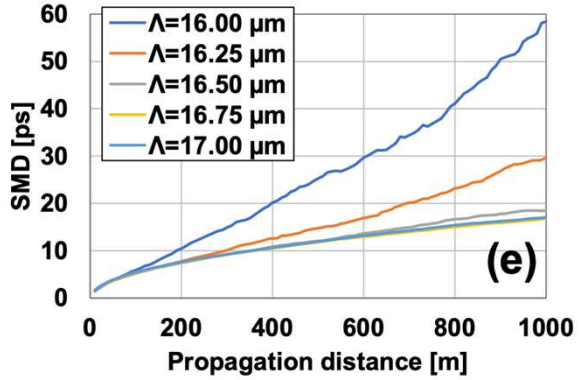


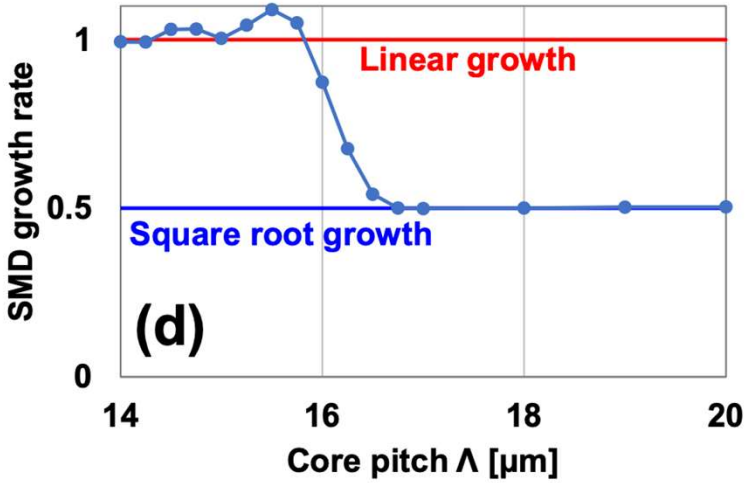


Geo
rg


(a)




Λ=16.00 µm
Λ=16.25 µm
Λ=16.50 µm
Λ=16.75 µm
Λ=17.00 µm
SMD [ps]
0
10
20
30
40
50
60
0
200
400
600
800
1000
Propagation distance [m]
(e)


SMD growth rate
0
0.5
1
Linear growth
Square root growth
(d)
14
16
18
20
Core pitch Λ [µm]

University of Stuttgart
Institute of electrical and optical communications

Thank you!



Prof. Dr.-Ing. Georg Rademacher

e-mail georg.rademacher@int.uni-stuttgart.de
phone +49 (0) 711 6856 7901
www. int.uni-stuttgart.de/

University of Stuttgart
Institute of electrical and optical communications

